Status Report
Week 8

JetRover Autonomous Retrieval

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Project Goal & Starting Point

- Goal: find an object, drive to it, pick it up, and bring it back to the starting location.
- Manual driving had previously worked on the robot.
- Existing code supported color detection, approach, and basic arm actions.
- This week focused on software; the robot was unavailable.

Work Completed This Week

- Reviewed the repository and robot driver interfaces.
- Implemented a mission that saves the starting position and heading.
- Added odometry-based return navigation and placement at home.
- Updated configuration, launch options, and operating instructions.

Updated Mission Flow

Save home > Search > Align > Approach > Pick
Confirm pickup > Return home > Place > Complete

- Camera detection tracks a red, green, or blue object.
- Depth readings guide alignment and approach to pickup range.
- The arm lifts the object into a carrying pose.
- The robot returns to its saved pose before placing it.

Control & Reliability Improvements

- Added explicit start/stop controls and mission status reporting.
- Added sensor freshness checks and movement timeouts.
- Removed forward movement without valid depth.
- Made arm actions responsive to stop requests; added gripper limits and a chassis command timeout.

Current Status & Limitations

- Initial implementation is complete; no builds or tests have been run.
- Pickup confirmation requires an operator by default.
- Automatic return can be enabled, but pickup then remains unverified.
- Return assumes a clear route. Obstacle avoidance and physical return accuracy remain unresolved.

Next Steps

- Install the mission package and updated chassis driver on the robot.
- Confirm sensor topics and camera/depth registration.
- Calibrate pickup distance, arm poses, and gripper settings.
- Check stopping and short return trips, then run a supervised retrieval mission.